



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING

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SECI1143 PROBABILITY & STATISTICAL DATA ANALYSIS

SECTION 02

Assignment 4

LECTURER: DR. NOORFA HASZLINNA BINTI MUSTAFFA

GROUP 4

Student Name	Matric No.
LAM YOKE YU	A23CS0233
GOE JIE YING	A23CS0224
TEH RU QIAN	A23CS0191
TAN YI YA	A23CS0187

Question 1

(a) Pearson's product-moment correlation coefficient and Spearman's rho rank correlation coefficient.

(b)

No.	x	y	xy	x ²	y ²
1.	26	42	1092	676	1764
2.	44	60	2640	1936	3600
3.	53	69	3657	2809	4761
4.	29	47	1363	841	2209
5.	77	91	7007	5929	8281
6.	80	98	7840	6400	9604
7.	20	39	780	400	1521
8.	40	55	2200	1600	3025
9.	67	85	5695	4489	7225
10.	86	104	8944	7396	10816
11.	17	37	629	289	1369
12.	61	77	4697	3721	5929
Σ	600	804	46544	36486	60104

$$r = \frac{\Sigma xy - \frac{\Sigma x \Sigma y}{n}}{\sqrt{\left(\Sigma x^2 - \frac{(\Sigma x)^2}{n}\right) \left(\Sigma y^2 - \frac{(\Sigma y)^2}{n}\right)}}$$

$$r = \frac{46544 - \frac{(600)(804)}{12}}{\sqrt{\left(36486 - \frac{(600)^2}{12}\right) \left(60104 - \frac{(804)^2}{12}\right)}}$$

$$= 0.998$$

Since $r = 0.998$, it has a very strong positive linear relationship between the number of items produced and the production cost.

Question 2

(a)

No.	x	y	xy	x ²	y ²
1.	0.27	2	0.54	0.0729	4
2.	1.41	3	4.23	1.9881	9
3.	2.19	3	6.57	4.7961	9
4.	2.83	6	16.98	8.0089	36
5.	2.19	4	8.76	4.7961	16
6.	1.81	2	3.62	3.2761	4
7.	0.85	1	0.85	0.7225	1
8.	3.05	5	15.25	9.3025	25
Σ	14.6	26	56.8	32.9632	104

$$r = \frac{\Sigma xy - \frac{\Sigma x \Sigma y}{n}}{\sqrt{(\Sigma x^2 - \frac{(\Sigma x)^2}{n})(\Sigma y^2 - \frac{(\Sigma y)^2}{n})}}$$

$$= \frac{56.8 - \frac{(14.6)(26)}{8}}{\sqrt{(32.9632 - \frac{(14.6)^2}{8})(104 - \frac{(26)^2}{8})}}$$

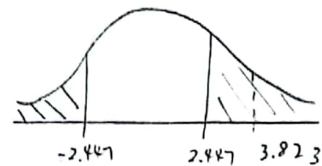
$$= 0.842$$

Since $r = 0.842$, it has a strong positive linear association between x and y .

(b) $H_0: \rho = 0$
 $H_1: \rho \neq 0$

Test statistic: $t = \frac{r}{\sqrt{\frac{1-r^2}{n-2}}} = \frac{0.842}{\sqrt{\frac{1-(0.842)^2}{8-2}}} = 3.823$

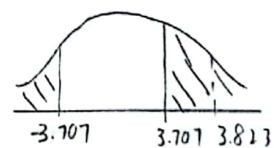
Critical value $\pm t_{0.025, 6} = \pm 2.447$



$\therefore H_0$ is rejected. There is sufficient evidence to conclude that there is linear correlation between size of household and weight of plastic usage.

(c) Critical value: $\pm t_{0.005, 6} = \pm 3.707$

$\therefore H_0$ is rejected. There is sufficient evidence to conclude that there is linear correlation between size of household and weight of plastic usage.



Subject :

No :

Date :

Question 3

(a) Engagement score and Sentiment score

(b) Likes and shares

(c)

Post ID	Engagement score	Sentiment Score	d_i	d_i^2
1	3	3	0	0
2	5	4.5	0.5	0.25
3	1	1.5	-0.5	0.25
4	6	6	0	0
5	2	1.5	0.5	0.25
6	4	4.5	-0.5	0.25

$$\sum d_i^2 = 1$$

$$r_s = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)}$$

$$= 1 - \frac{6 \times 1}{6(36 - 1)}$$

$$= 0.971$$

(d) (a) likes and comment

x	y	xy	x^2	y^2
150	20	3000	22500	400
100	10	1000	10000	100
200	25	5000	40000	625
80	8	640	6400	64
170	22	3740	28900	484
120	15	1800	14400	225
$\sum x = 820$	$\sum y = 100$	$\sum xy = 15180$	$\sum x^2 = 122200$	$\sum y^2 = 1898$

$$r = \frac{\sum xy - (\sum x \sum y) / n}{\sqrt{[(\sum x^2) - (\sum x)^2 / n][(\sum y^2) - (\sum y)^2 / n]}}$$

$$= \frac{15180 - (820 \times 100) / 6}{\sqrt{(122200 - 820^2 / 6)(1898 - 100^2 / 6)}}$$

$$= 0.988$$

(b) likes and share

x	y	xy	x ²	y ²
150	30	4500	22500	900
100	20	2000	10000	400
200	40	8000	40000	1600
80	15	1200	6400	225
170	35	5950	28900	1225
120	25	3000	14400	625
$\Sigma x = 820$	$\Sigma y = 165$	$\Sigma xy = 24650$	$\Sigma x^2 = 122200$	$\Sigma y^2 = 4975$

$$r = \frac{24650 - (820 \times 165) / 6}{\sqrt{(122200 - 820^2 / 6)(4975 - 165^2 / 6)}}$$

$$= 0.997$$

(c) Engagement score and Sentiment Score

x	y	xy	x ²	y ²
85	4	340	7225	16
70	3	210	4900	9
90	5	450	8100	25
60	2	120	3600	4
88	5	440	7744	25
75	3	225	5625	9
$\Sigma x = 468$	$\Sigma y = 22$	$\Sigma xy = 1785$	$\Sigma x^2 = 37194$	$\Sigma y^2 = 88$

$$r = \frac{1785 - (468 \times 22) / 6}{\sqrt{(37194 - 468^2 / 6)(88 - 22^2 / 6)}}$$

$$= 0.970$$

positive

In conclusion, the strongest correlation among (a), (b) and (c) are like and comment. Meanwhile, there are no negative correlation in (a), (b) and (c).

Subject:

No:

Date: . . .

$$(e) H_0: \rho = 0 \quad \alpha = 0.05, df = 6 - 2 = 4$$

$$H_1: \rho \neq 0 \quad t_{0.05, 4} = 2.776$$

x	y	xy	x ²	y ²
85	200	17000	7225	40000
70	100	7000	4900	10000
90	250	22500	8100	62500
60	150	9000	3600	22500
88	220	19360	7744	48400
75	180	13500	5625	32400
$\Sigma x = 468$	$\Sigma y = 1100$	$\Sigma xy = 88360$	$\Sigma x^2 = 37194$	$\Sigma y^2 = 215800$

$$r = \frac{88360 - (468 \times 1100) / 6}{\sqrt{(37194 - 468^2 / 6)(215800 - 1100^2 / 6)}}$$

$$= 0.820$$

$$\text{test statistic, } t = \frac{0.820}{\sqrt{\frac{1 - (0.820)^2}{4}}}$$

$$= 2.865$$

since $2.865 > t_{0.05, 4} = 2.776$, H_0 is rejected.

There is enough evidence to prove that linear correlation existed between engagement score and post length.

Question 4

(a)

	x	y	xy	x^2	y^2
	95	85	8075	9025	7225
	85	95	8075	7225	9025
	80	70	5600	6400	4900
	70	75	5250	4900	5625
	65	70	4550	4225	4900
Σ	395	395	31550	31775	31675

$$\begin{aligned}\therefore \Sigma x &= 395, \\ \Sigma y &= 395, \\ \Sigma xy &= 31550, \\ \Sigma x^2 &= 31775\end{aligned}$$

(b)

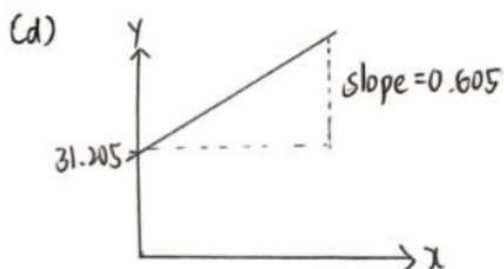
$$b_1 = \frac{\Sigma xy - \frac{\Sigma x \Sigma y}{n}}{\Sigma x^2 - \frac{(\Sigma x)^2}{n}}$$

$$= \frac{31550 - \frac{395 \times 395}{5}}{31775 - \frac{395^2}{5}}$$

$$= 0.605$$

$$\begin{aligned}b_0 &= \bar{y} - b_1 \bar{x} \\ &= 395/5 - 0.605(395/5) \\ &= 31.205\end{aligned}$$

(c) $\hat{y} = 31.205 + 0.605x$



(e) $\hat{y} = 31.205 + 0.605(60)$
 $= 67.505$

$\therefore$ grade C (67.505)

(f)

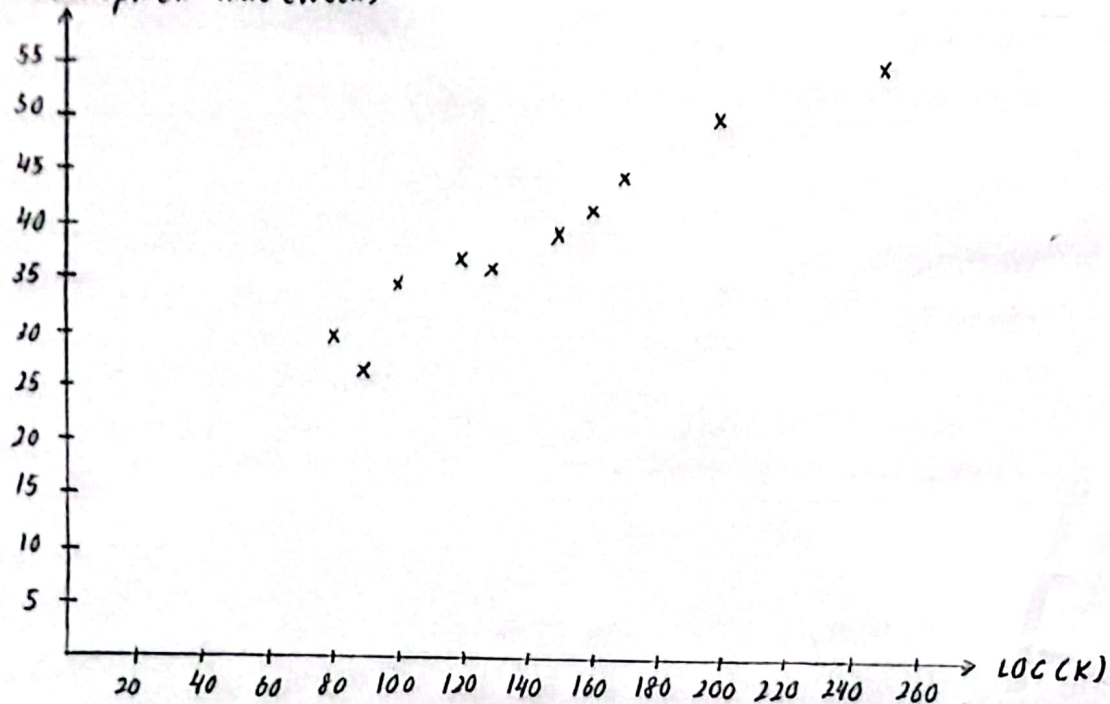
y	$\hat{y}$	$(\hat{y} - \bar{y})^2$	$(y - \bar{y})^2$
85	88.680	93.702	36
95	82.630	13.177	256
70	79.605	0.366	81
75	73.555	29.648	16
70	70.530	71.741	81
Σ		208.634	470

$$\begin{aligned}SST &= 470 \\ SSR &= 208.634 \\ R^2 &= \frac{SSR}{SST} \\ &= \frac{208.634}{470} \\ &= 0.444\end{aligned}$$

(g) good. 44.4% of the variation in statistics grade is explained by variation in score on math aptitude test.

Question 5

(a) Development Time (week)



(b)	ID	x	y	xy	x^2	y^2	$(y - \bar{y})^2$	$\hat{y}$	$(\hat{y} - \bar{y})^2$
	1	150	40	6 000	22 500	1 600	0	40.782	0.612
	2	100	35	3 500	10 000	1 225	25	32.977	49.323
	3	200	50	10 000	40 000	2 500	100	48.581	73.737
	4	80	30	2 400	6 400	900	100	29.855	102.921
	5	170	45	7 650	28 900	2 025	25	43.904	15.241
	6	120	38	4 560	14 400	1 444	4	36.099	15.218
	7	160	42	6 720	25 600	1 764	4	42.343	5.490
	8	90	28	2 520	8 100	784	144	31.416	73.685
	9	250	55	13 750	62 500	3 025	225	56.392	268.698
	10	130	37	4 810	16 900	1 369	9	37.660	5.476
	Σ	1630	400	61 910	267 700	16 636	636	45.465	610.399

For part (d)

$$\bar{y} = 40$$

$$\bar{x} = 163$$

$$r = \frac{\sum xy - \frac{\sum x \sum y}{n}}{\sqrt{\left(\sum x^2 - \frac{(\sum x)^2}{n}\right) \left(\sum y^2 - \frac{(\sum y)^2}{n}\right)}}$$

$$r = \frac{61910 - \frac{1450(400)}{10}}{\sqrt{\left(235300 - \frac{1450^2}{10}\right) \left(16636 - \frac{400^2}{10}\right)}}$$

$$r = 0.980$$

$$(c) b_1 = \frac{\sum xy - \frac{\sum x \sum y}{n}}{\sum x^2 - \frac{(\sum x)^2}{n}}$$

$$b_1 = \frac{61910 - \frac{1450(400)}{10}}{235300 - \frac{1450^2}{10}}$$

$$b_1 = 0.1561$$

$$b_0 = \bar{y} - b_1 \bar{x}$$

$$b_0 = 40 - 0.1561(163)$$

$$b_0 = 17.367$$

$$\hat{y}_i = 17.367 + 0.1561x$$

Interpretation:

b_0 is the estimated average development time when Line of codes (LOC) is zero. However, no software projects has zero lines. So for projects within the range of LOC observed, 17.367 weeks is the portion of development time not explained by LOC.

$b_1 = 0.1561$ shows that on average, the average value of development time increases by 0.1561 weeks for each additional 1K lines of code.

$$(d) \text{ SSR} = \sum (\hat{y} - \bar{y})^2$$

$$\text{SSR} = 610.399$$

$$\text{SST} = \sum (y - \bar{y})^2$$

$$\text{SST} = 636$$

$$R^2 = \frac{610.399}{636}$$

$$R^2 = 0.960$$

The high R^2 value indicates a strong linear relationship between development time and lines of code.

$$(e) \hat{y}_i = 17.367 + 0.1561x$$

$$x = 180$$

$$\hat{y} = 17.367 + 0.1561(180)$$

$$\hat{y} = 45.465$$